Arm Pose Robot — Quick Start

1. Start it

Run the script on the Pi. It prints the addresses it's serving on:

Stream: http://<pi-ip>:5000
Dashboard: http://<pi-ip>:5000/dashboard

Open the Dashboard link in a browser on your phone or laptop (same network).

2. What you see

Live feed with overlays:

- Skeleton drawn over the person being tracked.
- Status box top-right: `SEEK` (no one locked) → `LOCK` (tracking someone) → `GRACE` (briefly lost them, holding on).
- Lock box around the tracked person green = locked, orange = grace.
- Pose panel (right side): each arm's current best-guess pose, a confidence bar, and the last *confirmed* pose.
- Bottom bar: live shoulder/elbow angles for both arms.

Below the feed two cards, Right Arm and Left Arm. Each lists every pose with:

- An example count, colour-coded:
  - 🔴 red = under 5 (not enough)
  - 🟠 amber = 5–11 (works, but improve it)
  - 🟢 green = 12+ (reliable)
- + Record and Clear buttons.

3. Train a pose

The model learns by example — you show it a pose, then tell it what that pose is called.

1. Stand in front of the camera until the box turns green (LOCKED).
2. Hold the pose you want to teach (e.g. arm 90° up).
3. On the dashboard, tap + Record next to that pose's name.

4. Repeat 12+ times, shifting slightly each time step a bit nearer/further, angle yourself a little differently, this helps with reliability.
5. Count adjusts from red to green as examples are added.
6. Do the same for every pose you want the arm to recognise.

Minimum of 5 examples before a pose will ever be detected

> Tip: if the feed shows a `LOW VIS` warning, that arm isn't fully visible recordings will be skipped. Make sure the shoulder, elbow and wrist are all in frame.

4. Run it

Once trained, just hold a pose. The system needs the pose held steadily for about half a second (it filters out flickers), then the arm moves to mirror it. There's a short cooldown between moves so it won't twitch.

5. Check your training quality

- Tap Diagnose R / Diagnose L prints an accuracy report and confusion matrix to the Pi's console. It tells you exactly which poses are getting mixed up, so you know what to re-record.

6. Fixing / redoing

- Clear (next to a pose) — wipes just that pose so you can re-record it.
- Clear All — wipes everything (asks for confirmation).
- Training saves automatically and survives a restart, so you only calibrate once per setup.

Servo Test Tool — Quick Start

A terminal tool for scanning, testing, and configuring the HTD-85H bus servos.

Start it

Run the script. It connects to `/dev/ttyUSB0`, scans for servos, each found servo reports its ID, position, temperature, and voltage. USB bus may need to be changed for each dev board inserted.

Important: every servo needs a unique ID

All servos share one data line (a daisy-chained bus). They are addressed by ID, so two servos with the same ID will both answer at once and corrupt the bus causing errors or non detection.

Fix this by connecting servos one at a time and assigning each a unique ID with the `id` command before chaining them together:

id <current> <new>   e.g. id 1 3

After reassigning, power-cycle the servo, then `scan` to confirm.

Commands

Command	What it does
`scan`	Rescan the bus for servos
`status`	Show position / temp / voltage for all
`move <id> <angle> <ms>`	Move one servo (e.g. `move 1 90 1000`)
`move all <angle> <ms>`	Move every servo to the same angle
`joint <a1> <a2> <a3> <a4> <ms>`	Move each servo to its own angle (e.g. `joint 45 90 135 180 1000`)
`centre`	Send all servos to 120° (centre) over 1s
`stop <id>`	Stop one servo immediately

`stop all`	Stop all servos
`torque <id> on|off`	Enable/disable holding torque on one servo
`torque all on|off`	Enable/disable torque on all servos
`id <current> <new>`	Reassign a servo's ID (power-cycle after)
`help`	Reprint the command list
`quit`	Disable torque on all servos and exit

Limits

- **Angle:** 0–240°
- **Time:** 0–30000 ms
- **New ID:** 0–253

Handy to know

- `torque off` lets you move a servo by hand — useful for reading off pose angles with `status` while you position the arm manually.
- `quit` (not Ctrl-C) is the clean exit: it releases torque so nothing stays locked.